Friday 29 April 2022
02:00 pm – 04:00 pm
Duration: 2 hours
Additional time: 30 minutes
Timed exam – fixed start time

DEGREES OF MSc, MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

INFORMATION RETRIEVAL H
COMPSCI 4069

(Answer all questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks

1. **(a)** The following documents have been processed by an IR system where stemming is not applied:

DocID	Text
Doc1	france is world champion 1998 france won
Doc2	croatia and france played each other in the semifinal
Doc3	croatia was in the semifinal 1998
Doc4	croatia won the other semifinal in russia 2018

(i) Assume that the following terms are stopwords: and, in, is, the, was. Construct an inverted file for these documents, showing clearly the dictionary and posting list components. Your inverted file needs to store sufficient information for computing a simple tf*idf term weight, where $w_{ij} = tf_{ij} * \log_2(N/df_i)$

[5]

(ii) Compute the term weights of the two terms “champion” and “1998” in Doc1. Show your working.

[2]

(iii) Assuming the use of a best match ranking algorithm, rank all documents using their relevance scores for the following query:

1998 croatia

Show your working. Note that $\log_2(0.75)$= -0.4150 and $\log_2(1.3333)$= 0.4150.

[3]

(b)

(i) In Web search, explain why the use of raw term frequency (TF) counts in scoring documents can hurt the effectiveness of the search engine.

[2]

(ii) Suggest a solution to alleviate the problem, and show through examples how it might work. Explain through examples how modern term weighting models in IR control the raw term frequency counts.

[3]

(c) Assume that you have decided to modify the approach you use to rank the documents of your collection. You have developed a new Web ranking approach that makes use of recent advances in neural networks. All other components of the system remain the same. Explain in detail the steps you need to undertake to determine whether your new Web ranking approach produces a better retrieval performance than the original ranking approach.

[5]

2. **(a)** Consider a corpus of documents C written in English, where the frequency distribution of words approximately follows Zipf's law $r * p(w_r|C) = 0.1$, where $r = 1,2, \ldots, n$ is the rank of a word by decreasing order of frequency. w_r is the word at rank r, and $p(w_r|C)$ is the probability of occurrence of word w_r in the corpus C.

Compute the probability of occurrence of the most frequent word in C. Compute the probability of occurrence of the 2nd most frequent word in C. *Justify your answers*.

[4]

(b) Consider the query "michael jackson music" and the following term frequencies for the three documents D1, D2 and D3, where the search engine is using raw term frequency (TF) but no IDF:

	indiana	jackson	life	michael	music	pop	really
D1	0	4	1	3	0	6	1
D2	4	0	3	4	1	0	2
D3	0	4	0	5	4	4	0

Assume that the system has returned the following ranking: D2, D3, D1. The user judges D3 to be relevant and both D1 and D2 to be non-relevant.

(i) Show the original query vector, clearly stating the dimensions of the vector.

[2]

(ii) Use Rocchio's relevance feedback algorithm (with $\alpha=\beta=\gamma=1$) to provide a revised query vector for "michael jackson music". Terms in the revised query that have negative weights can be dropped, i.e. their weights can be changed back to 0. *Show all your calculations*.

[4]

(c) Suppose we have a corpus of documents with a dictionary of 6 words w_1, ..., w_6. Consider the table below, which provides the estimated language model $p(w|C)$ using the entire corpus of documents C (second column) as well as the word counts for doc_1 (third column) and doc_2 (fourth column), where $ct(w, doc_i)$ is the count of word w (i.e. its *term frequency*) in document doc_i. Let the query q be the following:

$$q = w_1\ w_2$$

Word	$p(w\|C)$	$ct(w, doc_1)$	$ct(w, doc_2)$
w_1	0.8	2	7
w_2	0.1	3	1
w_3	0.025	2	1
w_4	0.025	2	1
w_5	0.025	1	0
w_6	0.025	0	0
SUM	1.0	10	10
Word	$p(w\|C)$	$ct(w, doc_1)$	$ct(w, doc_2)$

(i) Assume that we do not apply any smoothing technique to the language model for doc_1 and doc_2. Calculate the query likelihood for both doc_1 and doc_2, i.e. $p(q|doc_1)$ and $p(q|doc_2)$ (Do not compute the log-likelihood; i.e. do not apply any log scaling). *Show your calculations.* Provide the resulting ranking of documents and state the document that would be ranked the highest.

[3]

(ii) Suppose we now smooth the language model for doc_1 and doc_2 using Jelinek-Mercer Smoothing with $\lambda = 0.1$. Recalculate the likelihood of the query for both doc_1 and doc_2, i.e., $p(q|doc_1)$ and $p(q|doc_2)$ (Do not compute the log-likelihood; i.e. do not apply any log scaling). *Show your calculations.* Provide the resulting ranking of documents and state the document that would be ranked the highest.

[4]

(iii) Explain which document you think should be reasonably ranked higher (doc_1 or doc_2) and why?

[3]

3. **(a)** Consider a query q for which there are 10 relevant documents in a corpus of 100 documents. Consider an IR system whose ranking for this query is shown below. In particular, its top 15 ranked documents were judged for relevance where R means the document was deemed relevant and N otherwise (the leftmost item is the top ranked search result).

System ranking: R N N R R N N N N R N N N N R

(i) Compute the precision, recall and the precision at rank 5 (P@5) of the system for this query. *Show your calculations.*

[3]

(ii) Compute the average precision (AP) of the system for this query. *Show your calculations.*

[2]

(b) A Web search engine receives a large number of ambiguous queries. Briefly discuss three possible techniques a Web search engine might use to deal with such queries and how they might help with ambiguous queries.

[3]

(c) Using an illustrative example, explain why Precision at Rank k (P@k) is not a good metric to use for learning an effective learning-to-rank model. What metric would you suggest to use instead and why?

[4]

(d) A posting list for a term in an inverted index contains the following three entries:

id=4 tf=3 id=6 tf=2 id=9 tf=4

Applying the delta compression of ids, show the binary form of the posting list when compressed using Unary compression for both ids and frequencies. What is the resulting (mean) compression rate, in bits per integer?

[3]

. **(e)** Neural retrieval models often use a re-ranking strategy over BM25 to reduce computational overhead.

Explain the key limitation of this strategy. Describe in sufficient details an approach that you might use to overcome this problem.

[5]